TASK 2 — Security Alert Monitoring & Incident Response

2.1 Objective

The objective of this task was to analyze system log files to detect suspicious activities such as brute force attacks, unauthorized login attempts, and abnormal network behavior.

2.2 Log Analysis

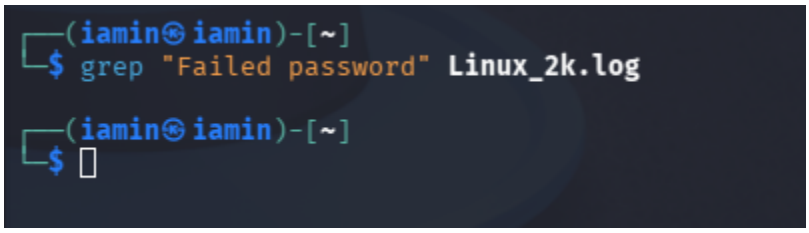
Linux system logs were analyzed to detect security events.

Example suspicious logs:

authentication failure user=root
rhost=207.243.167.114

This indicates repeated failed SSH login attempts.

📷 **Failed Login Detection**



2.3 SSH Activity Monitoring

SSH login attempts were analyzed to detect brute force patterns.

📷 **SSH Activity Screenshot**

```
┌──(iamin㉿iamin)-[~]
└─$ grep -i "ssh" Linux_2k.log
Jun 14 15:16:01 combo sshd(pam_unix)[19939]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=218.188.2.4
Jun 14 15:16:02 combo sshd(pam_unix)[19937]: check pass; user unknown
Jun 14 15:16:02 combo sshd(pam_unix)[19937]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=218.188.2.4
Jun 15 02:04:59 combo sshd(pam_unix)[20882]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=220-135-151-1.hinet-ip.hinet.net  user=root
Jun 15 02:04:59 combo sshd(pam_unix)[20884]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=220-135-151-1.hinet-ip.hinet.net  user=root
Jun 15 02:04:59 combo sshd(pam_unix)[20883]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=220-135-151-1.hinet-ip.hinet.net  user=root
Jun 15 02:04:59 combo sshd(pam_unix)[20885]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=220-135-151-1.hinet-ip.hinet.net  user=root
Jun 15 02:04:59 combo sshd(pam_unix)[20886]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=220-135-151-1.hinet-ip.hinet.net  user=root
Jun 15 02:04:59 combo sshd(pam_unix)[20892]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=220-135-151-1.hinet-ip.hinet.net  user=root
Jun 15 02:04:59 combo sshd(pam_unix)[20893]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=220-135-151-1.hinet-ip.hinet.net  user=root
Jun 15 02:04:59 combo sshd(pam_unix)[20896]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=220-135-151-1.hinet-ip.hinet.net  user=root
Jun 15 02:04:59 combo sshd(pam_unix)[20897]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=220-135-151-1.hinet-ip.hinet.net  user=root
Jun 15 02:04:59 combo sshd(pam_unix)[20898]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=220-135-151-1.hinet-ip.hinet.net  user=root
Jun 15 12:12:34 combo sshd(pam_unix)[23397]: check pass; user unknown
Jun 15 12:12:34 combo sshd(pam_unix)[23397]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=218.188.2.4
Jun 15 12:12:34 combo sshd(pam_unix)[23395]: check pass; user unknown
Jun 15 12:12:34 combo sshd(pam_unix)[23395]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=218.188.2.4
Jun 15 12:12:34 combo sshd(pam_unix)[23404]: check pass; user unknown
Jun 15 12:12:34 combo sshd(pam_unix)[23404]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=218.188.2.4
Jun 15 12:12:34 combo sshd(pam_unix)[23399]: check pass; user unknown
Jun 15 12:12:34 combo sshd(pam_unix)[23399]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=218.188.2.4
Jun 15 12:12:34 combo sshd(pam_unix)[23406]: check pass; user unknown
Jun 15 12:12:34 combo sshd(pam_unix)[23406]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=218.188.2.4
Jun 15 12:12:34 combo sshd(pam_unix)[23396]: check pass; user unknown
Jun 15 12:12:34 combo sshd(pam_unix)[23394]: check pass; user unknown
Jun 15 12:12:34 combo sshd(pam_unix)[23407]: check pass; user unknown
Jun 15 12:12:34 combo sshd(pam_unix)[23394]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=218.188.2.4
Jun 15 12:12:34 combo sshd(pam_unix)[23403]: check pass; user unknown
Jun 15 12:12:34 combo sshd(pam_unix)[23396]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=218.188.2.4
Jun 15 12:12:34 combo sshd(pam_unix)[23407]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=218.188.2.4
Jun 15 12:12:34 combo sshd(pam_unix)[23403]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=218.188.2.4
Jun 15 12:12:34 combo sshd(pam_unix)[23412]: check pass; user unknown
```

2.4 Suspicious IP Analysis

Log data was processed using command line analysis to identify IP addresses with the highest number of login attempts.

Example command used:

grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' Linux_2k.log | sort | uniq -c | sort -nr

📷 **Suspicious IP Analysis**

```
┌──(iamin㉿iamin)-[~]
└─$ grep -Eo '([0-9]{1,3}\.){3}[0-9]{1,3}' Linux_2k.log | sort | uniq -c | sort -nr | head
     80 150.183.249.110
     46 62.99.164.82
     46 207.30.238.8
     46 203.101.45.59
     46 163.27.187.39
     46 059.45.101.203
     44 211.72.151.162
     41 211.107.232.1
     35 202.82.200.188
     32 83.116.207.11

┌──(iamin㉿iamin)-[~]
└─$
```

2.5 Login Attempts Detection

Multiple failed login attempts from the same IP address indicate possible brute force attacks.

📷 **Login Attempts Screenshot**

```
┌──(iamin㉿iamin)-[~]
└─$ grep -i "login" Linux_2k.log
Jun 30 20:53:04 combo klogind[19272]: Authentication failed from 163.27.187.39 (163.27.187.39): Permission denied i
n replay cache code
Jun 30 20:53:04 combo klogind[19272]: Kerberos authentication failed
Jun 30 20:53:04 combo klogind[19287]: Authentication failed from 163.27.187.39 (163.27.187.39): Permission denied i
n replay cache code
Jun 30 20:53:04 combo klogind[19287]: Kerberos authentication failed
Jun 30 20:53:04 combo klogind[19286]: Authentication failed from 163.27.187.39 (163.27.187.39): Permission denied i
n replay cache code
Jun 30 20:53:04 combo klogind[19286]: Kerberos authentication failed
Jun 30 20:53:04 combo klogind[19271]: Authentication failed from 163.27.187.39 (163.27.187.39): Permission denied i
n replay cache code
Jun 30 20:53:04 combo klogind[19271]: Kerberos authentication failed
Jun 30 20:53:04 combo klogind[19270]: Authentication failed from 163.27.187.39 (163.27.187.39): Permission denied i
n replay cache code
Jun 30 20:53:04 combo klogind[19270]: Kerberos authentication failed
Jun 30 20:53:04 combo klogind[19269]: Authentication failed from 163.27.187.39 (163.27.187.39): Permission denied i
n replay cache code
Jun 30 20:53:04 combo klogind[19269]: Kerberos authentication failed
Jun 30 20:53:04 combo klogind[19268]: Authentication failed from 163.27.187.39 (163.27.187.39): Permission denied i
n replay cache code
Jun 30 20:53:04 combo klogind[19288]: Authentication failed from 163.27.187.39 (163.27.187.39): Permission denied i
n replay cache code
Jun 30 20:53:04 combo klogind[19274]: Authentication failed from 163.27.187.39 (163.27.187.39): Software caused con
nection abort
Jun 30 20:53:04 combo klogind[19268]: Kerberos authentication failed
Jun 30 20:53:04 combo klogind[19288]: Kerberos authentication failed
Jun 30 20:53:04 combo klogind[19274]: Kerberos authentication failed
Jun 30 20:53:06 combo klogind[19266]: Authentication failed from 163.27.187.39 (163.27.187.39): Software caused con
nection abort
Jun 30 20:53:06 combo klogind[19266]: Kerberos authentication failed
Jun 30 20:53:06 combo klogind[19267]: Authentication failed from 163.27.187.39 (163.27.187.39): Software caused con
nection abort
Jun 30 20:53:06 combo klogind[19267]: Kerberos authentication failed
Jun 30 20:53:06 combo klogind[19278]: Authentication failed from 163.27.187.39 (163.27.187.39): Software caused con
nection abort
Jun 30 20:53:06 combo klogind[19278]: Kerberos authentication failed
Jun 30 20:53:06 combo klogind[19273]: Authentication failed from 163.27.187.39 (163.27.187.39): Software caused con
nection abort
Jun 30 20:53:06 combo klogind[19273]: Kerberos authentication failed
Jun 30 20:53:06 combo klogind[19276]: Authentication failed from 163.27.187.39 (163.27.187.39): Software caused con
nection abort
Jun 30 20:53:06 combo klogind[19276]: Kerberos authentication failed
Jun 30 20:53:06 combo klogind[19275]: Authentication failed from 163.27.187.39 (163.27.187.39): Software caused con
nection abort
Jun 30 20:53:06 combo klogind[19275]: Kerberos authentication failed
Jun 30 20:53:06 combo klogind[19277]: Authentication failed from 163.27.187.39 (163.27.187.39): Software caused con
nection abort
Jun 30 20:53:06 combo klogind[19277]: Kerberos authentication failed
Jun 30 20:53:06 combo klogind[19279]: Authentication failed from 163.27.187.39 (163.27.187.39): Software caused con
nection abort
```

📷 Authentication Log Detection

```
Jul 24 08:31:57 combo sshd(pam_unix)[22185]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=203.251.225.101  user=root
Jul 24 08:31:57 combo sshd(pam_unix)[22184]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=203.251.225.101  user=root
Jul 24 08:31:59 combo sshd(pam_unix)[22188]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=203.251.225.101  user=root
Jul 24 08:31:59 combo sshd(pam_unix)[22189]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=203.251.225.101  user=root
Jul 24 08:31:59 combo sshd(pam_unix)[22191]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=203.251.225.101  user=root
Jul 26 07:02:27 combo sshd(pam_unix)[28842]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:02:35 combo sshd(pam_unix)[28844]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:02:37 combo sshd(pam_unix)[28846]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:02:45 combo sshd(pam_unix)[28848]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:02:47 combo sshd(pam_unix)[28850]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:02:55 combo sshd(pam_unix)[28852]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:02:57 combo sshd(pam_unix)[28854]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:03:05 combo sshd(pam_unix)[28856]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:03:07 combo sshd(pam_unix)[28858]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:03:15 combo sshd(pam_unix)[28860]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:03:17 combo sshd(pam_unix)[28862]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:03:26 combo sshd(pam_unix)[28864]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:03:27 combo sshd(pam_unix)[28866]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:03:35 combo sshd(pam_unix)[28868]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:03:37 combo sshd(pam_unix)[28870]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:03:45 combo sshd(pam_unix)[28872]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:03:47 combo sshd(pam_unix)[28874]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:03:55 combo sshd(pam_unix)[28876]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:03:57 combo sshd(pam_unix)[28878]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:04:02 combo sshd(pam_unix)[28880]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:04:05 combo sshd(pam_unix)[28882]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:04:07 combo sshd(pam_unix)[28884]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root
Jul 26 07:04:12 combo sshd(pam_unix)[28886]: authentication failure; logname= uid=0 euid=0 tty=NODEVssh ruser= rhos
t=207.243.167.114  user=root

┌──(iamin㉿iamin)-[~]
└─$
```

2.6 Incident Response Actions

Recommended mitigation steps:

1. Block malicious IP addresses at firewall level

2. Implement login attempt rate limiting
3. Enable account lockout policies
4. Monitor logs continuously using SIEM tools

2.7 Task 2 Summary

Security incidents detected:

Incident	Severity
Brute Force Login Attempts	High
Unauthorized SSH Access Attempts	High
Suspicious IP Activity	Medium

GitHub Repository:

FUTURE_CS_02